100

CHE 321 - HW4

1 (a) $A + B \leftrightarrow C$ liquid

$C_{A0} = C_{B0} = 2M$

$K_c = 10\, L/mol$

$$K_c = \frac{C_C}{C_A C_B} = 10\, L/mol = \frac{C_{A0} X}{C_{A0}(1-x)\, C_{A0}(1-x)} = \frac{C_{A0} x}{C_{A0}^2 (1-x)^2} = \frac{x}{C_{A0}(1-x)^2} = 10$$

$C_A = C_{A0}(1-x)$

$C_B = C_{B0}(1-x) = C_{A0}(1-x) = C_A$

$C_C = C_{A0} X$

$x = 10 C_{A0}(1-x)^2$

$x = 10(2M)(1-x)^2$

$x = 20(1 - 2x + x^2)$

$$\frac{x}{1-2x+x^2} = 20$$

$\boxed{x = .8}$

$C_A = C_{A0}(1-x)$

$C_A = (2M)(1-.8)$

$\boxed{C_A = .4M}$ $C_A = C_B$

$\boxed{C_B = .4M}$

$C_C = C_{A0} x \Rightarrow C_C = (2M)(.8) \Rightarrow \boxed{C_C = 1.6M}$

(b) $A \leftrightarrow 3C$ gas PFR (no P drop)

$T_{A0} = 400K$ $K_c = .25\, L^2/mol^2$

$P_{A0} = 10\, atm$

$A \leftrightarrow$	$3C$
V_0	0
$V_0 X$	$3 V_0 X$
$V_0(1-x)$	$3 V_0 X$

$V_0(1-x) + 3V_0 x = V_0(1+2x) = V_t$

$PV = NRT$

$$\frac{P}{RT} = \frac{N}{V}$$

$$C_A = \frac{N_A}{V_t} = \frac{N_{A0}(1-x)}{V_0(1+2x)} = \frac{C_{A0}(1-x)}{(1+2x)}$$

$$C_{A0} = \frac{P_0}{RT} = \frac{10\, atm}{(.08206)(400K)} = .305$$

$$K_c = \frac{C_C^3}{C_A} = \frac{\left(3 C_{A0} \frac{x}{1+2x}\right)^3}{C_{A0}\left(\frac{1-x}{1+2x}\right)} = .25 = \frac{(9)(.305)^3\left(\frac{x}{1+2x}\right)^3}{.305\left(\frac{1-x}{1+2x}\right)} \Rightarrow \boxed{x = .58}$$

$$C_A = \frac{C_{A0}(1-x)}{1+2x} = \frac{(.305)(1-.58)}{1+2(.58)} \Rightarrow \boxed{C_A = .059}$$

$$C_C = 3C_{A0}\left(\frac{x}{1+2x}\right) = 3(.305)\left(\frac{.58}{1+2(.58)}\right) \Rightarrow \boxed{C_C = .25}$$

② $A \rightarrow B$ $\qquad$ Production rate $= 1000\ \frac{lb}{hr} = F_B$

$k = .0015\ min^{-1}$ $\qquad$ $E_a = 25000\ \frac{cal}{mol}$

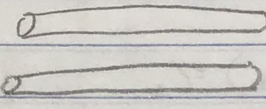
$T = 80°F = 299.67K$ $\qquad$ $X_A = .90$ $\qquad$ $V_{pipe} = \pi r^2 L$

Tubes: $L = 10ft$ $\qquad$ $PV = nRT$ $\qquad$ $V_{pipe} = \pi\left(\left(\frac{1}{12}/2\right)\right)^2 \cdot 10ft$

$d = 1in$ $\qquad$ $MW_A = MW_B = 58$ $\qquad$ $V_{pipe} = .0545\ ft^3$

$P_p = 132\ psig$

$T_p = 260°F = 399.67K$



PFRs in parallel

$$\frac{dF_A}{dV} = -r_A = kC_{A0}(1-x) \qquad C_{A0} = \frac{N_0}{V_t} = \frac{P}{RT} = \frac{(132\ psig + 14.7)\ psia}{(10.73)(719.67R)}$$

$F_B = F_{A0}X$ $\qquad$ $F_{A0} =$ $\qquad$ $C_{A0} = .0189\ \frac{lbmol}{ft^3}$

$$F_{A0} = \frac{F_B}{X} = \frac{1000\ lb/h}{.90} = 1111.1\ lb/hr = 8689.6\ \frac{mol}{hr} \qquad C_{A0} = 8.57\ \frac{mol}{ft^3}$$

$K = Ae^{-Ea/RT}$

$$\ln\left(\frac{k_2}{k_1}\right) = \frac{Ea}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right)$$

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$$\ln\left(\frac{k_2}{.0015}\right) = \frac{25000}{(1.985\ \frac{cal}{K\,mol})}\left(\frac{1}{299.7K} - \frac{1}{399.7K}\right) \Rightarrow K_2 = 56.82\ min^{-1}$$

$$K_2 = 3409.2\ hr^{-1}$$

$$V_{tot} = \frac{F_{A0}}{k_2 C_{A0}}(-\ln(.1)) = \frac{(8689.6\ \frac{mol}{hr})(-\ln(.1))}{(3409.2\ hr^{-1})(8.57\ \frac{mol}{ft^3})}$$

$V_{tot} = .685\ ft^3$

$V_{tot} = n_{tubes} V_{single\ tube}$

$$n_{tubes} = \frac{V_{tot}}{V_{single\ tube}} = \frac{.685 ft^3}{.0545 ft^3} \Rightarrow n_{tubes} = 12.57$$

$$\boxed{n_{tubes} = 13}$$